

Financial Technology

HW1: K-Bar Deadline: 2025/10/13 23:59

Problem:

This dataset is the trading record of Taiwan Future Exchange (TAIFEX). The student should filter “TX”, “expired in Sep”, “8:45-13:45” and plot the bar learned from the lesson. The results should be included in report.

1. Save the filtered dataset as “TX.csv”. The date, time, price, and volume should be included.
2. Plot the time bar in the period of “1 Day”.
3. Plot the tick bar in the number of “10k” ticks.
4. Plot the volume bar in the volume of “100k”.
5. Plot the dollar bar every “1B” NTD.

Remark:

1. The color of bar is not limited, but you must explain it.
2. X and y axis are respectively timestamp and price, **the number of ticks** needs to be shown at the bottom.
3. **Any blank is not allowed.**

Time bar



Blank is not allowed

Grading Policy:

1. In the programming assignment, the **code(.ipynb or .py)**, **report(.pdf)**, **TX.csv**, and **Readme** file should all be compressed into a **ZIP** file named 學號_姓名 and upload to eeclass website. If your code contains external numerical or image data etc. instead of data stream, it should be contained in the ZIP too. The Readme file should explain how to run your code.
2. The programming language that can be used on this assignment is Python. Built-in libraries or functions are allowed to be used. Please integrate all your code into a main file.
3. According to the question requirement, the result should be contained in your **report**. **Completeness of responses will affect your score (Explanation, Process, Results, and Discussion).**
4. 30% off for late submission within one week, not accepted after one week.
5. Discussions are encouraged, **but plagiarism is strictly prohibited.**

